

Grade 12 Computer Science (083) — Worksheet - 2

Section A — Objective Type 1 MARK EACH

1. Text File Handling in Python

Q1. State if the following statement is True or False: The `readlines()` method reads all lines of a text file and returns them as a list of strings, where each string ends with a newline character.

Q2. Which method is used to write a single string to a text file in Python?

a) `write()`

b) `writeline()`

c) `writelines()`

d) `print()`

Q3. Consider the following code:

```
f = open("notes.txt", "a")
f.write("Chapter 1")
f.close()
```

What is the mode of the file `notes.txt` after this code is executed?

a) Read mode

b) Write mode

c) Append mode

d) Read and Write mode

2. Binary Files & CSV Handling

Q4. Which of the following modes will open a file for reading and writing in binary format, placing the file pointer at the beginning of the file without truncating it?

a) `wb+`

b) `rb+`

c) `ab+`

d) `wb`

Q5. What will be the output of the following Python code?

```
import csv
with open('marks.csv', 'w', newline='') as f:
    writer = csv.writer(f)
    writer.writerow(['Subject', 'Marks'])
    writer.writerow(['Physics', 88])
    writer.writerow(['Chemistry', 92])
```

- a) The code will create a CSV file named 'marks.csv' with three rows.
- b) The code will create a CSV file named 'marks.csv' with two rows.
- c) The code will raise an error because the file is opened in write mode.
- d) The code will append data to an existing 'marks.csv' file.

Q6. Which method of the `pickle` module is used to write a Python object to a binary file?

- a) `pickle.write()`
- b) `pickle.dump()`
- c) `pickle.store()`
- d) `pickle.save()`

3. Data Structures — Stacks

Q7. In the context of a stack, the operation to remove an element is known as:

- a) Push
- b) Pop
- c) Delete
- d) Remove

Q8. A stack is implemented using a list `stack = []`. Which of the following correctly implements the pop operation?

- a) `stack.pop()`
- b) `stack.remove()`
- c) `stack.delete()`
- d) `stack.discard()`

Q9. Consider the following implementation of a stack using a list:

```
stack = [5, 15, 25]
```

Write the Python code to push the element 35

onto the stack and then display the top element without removing it.

4. Relational Database & SQL — Basic Concepts

Q10. In a relational database, a row of a table is referred to as a/an:

a) Attribute

b) Tuple

c) Domain

d) Degree

Q11. In SQL, which constraint ensures that a column cannot have NULL values?

a) UNIQUE

b) PRIMARY KEY

c) NOT NULL

d) Both b) and c)

Q12. Which SQL keyword is used to sort the result set in ascending or descending order?

a) GROUP BY

b) SORT BY

c) ORDER BY

d) ARRANGE BY

5. SQL — DDL, DML & Clauses

Q13. Which of the following is the correct syntax to remove a table named 'CUSTOMER' from a database in MySQL?

a) REMOVE TABLE CUSTOMER;

b) DELETE TABLE CUSTOMER;

c) DROP TABLE CUSTOMER;

d) ERASE TABLE CUSTOMER;

Q14. In SQL, which clause is used to filter records before grouping?

a) HAVING

b) WHERE

c) GROUP BY

d) ORDER BY

Q15. What is the cardinality of a relation that has 6 attributes and 15 tuples?

a) 6

b) 15

c) 21

d) 90

Section B — Short Answer Type

2 MARKS EACH

6. Text File Methods

Q16. Differentiate between the `write()` and `writelines()` methods for writing data to a text file.

Q17. Write the output of the following Python code:

```
with open("poem.txt", "w") as f:  
    f.write("Twinkle twinkle little star how I wonder
```

```
what you are.")
with open("poem.txt", "r") as f:
    f.seek(8)
    data = f.read(7)
    print(data)
```

Q18. Write a Python statement to import the `pickle` module and then a statement to write an object `student` to a binary file named `student.dat` .

7. Relational Database — Keys & SQL DDL

Q19. Explain the concept of a candidate key and an alternate key in a relational database with a suitable example.

Q20. Write a SQL query to create a table named LIBRARY with the following attributes:

- Book_ID (Integer, Primary Key)
- Title (String of size 100, Not Null)
- Price (Float)

Q21. Consider the following implementation of a stack using a list:

```
stack = [100, 200, 300] . Write the Python code for the pop
```

operation, which removes and returns the top element from the stack, and displays "Stack Underflow" if the stack is empty.

Section C — Short Answer Type 3 MARKS EACH

8. File Handling — Text File Operations

Q22. Write a Python function named `count_is()` that reads the contents of a text file "ESSAY.TXT" and displays the number of times the word "is" (case-insensitive) appears in the file.

Q23. Write a Python function `save_marks()` that accepts a list of lists `data` (where each inner list is a record with fields [Roll_No, Name, Marks]) and writes it to a CSV file named `marks.csv` .

Q24. Write a Python program that reads a text file "SOURCE.TXT" and creates a new file "SHORT.TXT" containing only the lines from "SOURCE.TXT" that have less than 20 characters. Also, count and display the total number of lines written to "SHORT.TXT".

9. SQL — Queries with Conditions & Aggregation

Q25. Write a SQL query for the following based on a table INVENTORY with columns Item_ID, Item_Name, Price, Quantity:

- I. To display the item names and their prices for items whose quantity is between 20 and 80.
- II. To display the total price of all items.
- III. To display the minimum price for each item category (Assume a column named Category exists).

Q26. Consider two tables MOVIES and DIRECTORS:

Table: MOVIES

Movie_ID	Title	Director_ID	Genre	Rating
M101	Inception	D01	Sci-Fi	8.8
M102	Interstellar	D01	Sci-Fi	8.6
M103	Titanic	D02	Romance	7.9
M104	The Dark Knight	D03	Action	9.0
M105	Pulp Fiction	D04	Crime	8.9
M106	Dunkirk	D01	War	7.8
M107	Oppenheimer	D05	Biography	8.5

Table: DIRECTORS

Director_ID	Director_Name	Country
D01	Christopher Nolan	UK
D02	James Cameron	Canada

D03	Christopher Nolan	UK
D04	Quentin Tarantino	USA
D05	Christopher Nolan	UK

Write SQL queries for the following:

- I. To display the Title, Director_Name, and Rating of all Sci-Fi movies.
- II. To display the Country and the total number of movies directed by directors from each country.
- III. To display the Title and Director Name of movies that have a rating greater than the average rating of all movies.
- IV. To display the Title, Director Name, and Rating of movies sorted by Director Name in ascending order and Rating in descending order.

Q27. Consider two tables PLAYERS and MATCHES:

Table: PLAYERS

Player_ID	Player_Name	Team	City
201	Rohit Verma	India	Delhi
202	Karan Mehta	India	Mumbai
203	Arjun Nair	India	Kochi
204	Sameer Khan	India	Lucknow
205	Dev Patel	India	Ahmedabad

Table: MATCHES

Player_ID	Format	Runs	Match_Date
201	ODI	85	2025-02-10
201	T20	45	2025-02-12
202	ODI	102	2025-02-10
203	Test	150	2025-02-11

204	T20	30	2025-02-12
205	ODI	78	2025-02-10
202	T20	55	2025-02-12

Write SQL queries for the following:

- I. To display the Cartesian Product of the PLAYERS and MATCHES tables.
- II. To display the Player_Name, Format, and Runs of all players using an equi-join.
- III. To display the Player_Name, Format, and Runs of players who have scored more than 80 in any format.
- IV. To display the Player_Name and the average runs for each player, but only for those players whose average runs is greater than 60.

Section D — Long Answer Type

4 MARKS EACH

10. File Handling — CSV & Text File Integration

Q28. Consider a CSV file "STOCK.csv" with the following columns: Item_ID, Item_Name, Quantity, Price. Write Python functions to:

- I. `add_item()` — Accept the Item_ID, Item_Name, Quantity, and Price from the user and add the record to the end of the "STOCK.csv" file.
- II. `total_inventory_value()` — Calculate and return the total value of all items in stock (value = Quantity * Price).

Q29. Write a Python program that reads a text file "INPUT.TXT" and creates a new file "UPPER.TXT" containing only the lines from "INPUT.TXT" that start with an uppercase letter. Also, count and display the total number of lines written to "UPPER.TXT".

Q30. Write a Python function `save_employees()` that accepts a list of lists `data` (where each inner list is a record with fields [Emp_ID, Name, Department]) and writes it to a CSV file named `employees.csv`.

11. Stacks — Implementation & Applications

Q31. A list of products is given as:

```
products = [("P1", "Laptop", 55000), ("P2", "Mouse", 800),  
            ("P3", "Monitor", 12000), ("P4", "Keyboard", 1500)]
```

Write the following user-defined functions to perform operations on a stack named `Prod_Stack` :

- I. `Push_prod()` — To push the records of products whose price is greater than 5000 onto the stack.
- II. `Pop_prod()` — To pop the items from the stack and display them. Also, display 'Stack is Empty' when there are no elements in the stack.

Q32. Ms. Kavita, the librarian of a school, needs to maintain a stack of issued books. Each record includes: Book_ID, Book_Title, Student_Name, Issue_Date, Due_Date.

Write the Python functions to:

- I. `push_issue()` — Input book issue data from the user and push it onto a stack (implemented using a list).
- II. `pop_issue()` — Pop the top issued book record from the stack and display it.
- III. `display_issues()` — Display all the issued books present in the stack without removing them.

Q33. Consider the following implementation of a stack using a list:

```
stack = [11, 22, 33]
```

. Write the Python code for the pop operation,

which removes and returns the top element from the stack, and displays "Stack Underflow" if the stack is empty. Also, write a function to display all elements of the stack without removing them.

Section E — Case Study / Application Based

5 MARKS EACH

12. File Handling — Log File Analysis

Q34. Rohan, a system administrator, is working on a large text file "SERVER.log" that contains server log entries. Each line in the file has a timestamp and a message. Help him create the following user-defined functions:

- I. `count_critical_lines()` — to count and return the number of lines in the "SERVER.log" file that contain the word "CRITICAL".
- II. `create_debug_report()` — to read the "SERVER.log" file and write all lines that contain the word "DEBUG" to a new file "DEBUG.txt".
- III. `find_text_position()` — to take a text as an argument and find the first occurrence of the text in the "SERVER.log" file using the `seek()` and `tell()` methods, printing the position of the text if it is found.

Q35. Write a Python program that reads a text file "SERVER.log" and:

- I. Counts the total number of lines in the file.
- II. Counts the number of lines containing the word "CRITICAL".
- III. Creates a new file "CRITICAL.txt" containing only the lines with "CRITICAL".
- IV. Displays the first 60 characters of the file using `seek()` and `read()`.

Q36. A text file "DIARY.txt" contains personal diary entries. Write a Python program to:

- I. Count the number of words in the file.
- II. Count the number of lines that end with a full stop ('.').
- III. Create a new file "FULLSTOP.txt" with lines that end with a full stop.
- IV. Display the last 30 characters of the file.

13. SQL — Employee Database Queries

Q37. Consider the following table STAFF:

Staff_ID	Staff_Name	Department	Salary	Joining_Date
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S101	Manish Gupta	IT	68000	2019-06-10
S102	Neha Kapoor	HR	52000	2020-02-15
S103	Rahul Jain	Finance	71000	2018-12-01
S104	Pooja Mishra	IT	85000	2017-09-20
S105	Aditya Rao	Marketing	58000	2021-04-05
S106	Kritika Sen	IT	92000	2019-11-11
S107	Varun Das	HR	47000	2022-01-25

Predict the output of the following SQL queries:

I.

```
SELECT Staff_Name, Salary FROM STAFF WHERE Department = 'IT';
```

II.

```
SELECT Department, AVG(Salary) FROM STAFF GROUP BY Department;
```

III.

```
SELECT * FROM STAFF WHERE Staff_Name LIKE 'K%';
```

IV.

```
SELECT COUNT(*) FROM STAFF WHERE Joining_Date > '2019-01-01';
```

V.

```
SELECT Department, COUNT(*) AS Total_Staff, SUM(Salary) AS Total_Salary FROM STAFF GROUP BY Department;
```

Q38. Based on the STAFF table above, write SQL queries to:

- I. Display the names of staff whose salary is greater than the average salary of all staff.
- II. Display the department-wise minimum salary.
- III. Display the names and joining dates of staff who joined after 2019.
- IV. Display the total number of staff in each department, sorted by department name in descending order.
- V. Display the name and salary of the lowest-paid staff member.

Q39. Consider the STAFF table above. Write SQL queries to:

- I. Display the names of staff whose names contain the letter 'a' (case-insensitive).
- II. Display the average salary of the Finance department.
- III. Display the names and salaries of staff sorted by salary in ascending order.
- IV. Display the department and the number of staff who joined before 2020-01-01.
- V. Display the names of staff whose salary is between 50000 and 80000.